

ROBOPSYCHOLOGY

What we mean by Robopsychology

Robopsychology is about the well being of humans who are interacting with robots or AI-systems and about the effects of this interacting to those persons.

- putting human experience and needs at the center of technological development
- applying scientific methods and theories in psychology to current questions of AI and Robotics
- creating evidence-based guidelines for human-centered technology

Different perspectives on human-machine relations

An interplay of different factors determines the individual experience and acceptance of intelligent machines.

- agent-specific factors: visual appearance, behavior, communication style
- user-specific factors: age, personality factors, technical pre-experience
- contextual factors: area of application, information provided to new users

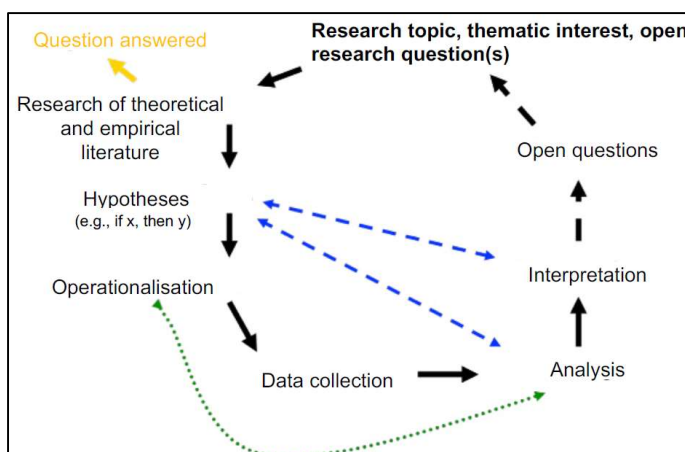
Definition of Psychology

Psychology is the study of the human mind and human behavior. It is a diverse scientific discipline comprising several major branches of research as well as several subareas of applied psychology.

Research in psychology involves observation, experimentation, testing, and analysis to explore the biological, cognitive, emotional, personal, and social processes underlying human behavior.

The focus of empirical research is on qualitative (e. g. interview), quantitative as well as statistical data and psychology research take place inside of laboratories, in the field (natural setting) or online.

The Scientific Cycle



Different types of data collection

- Population-descriptive:
Univariate (only one variable is measured), without underlying hypothesis ...
- Correlational study:
Bivariate, both variables are ordinally scaled, causal interpretation not allowed ...
- Experimental study:
One or more variables (independent variable), systematically varied to observe the effect on one or more other variables (dependent variable), when causal effects are to be tested ...
- Cross-sectional study:
In a cross-sectional study, an empirical study is conducted only once.
- Longitudinal study:
In a longitudinal study, the same empirical study is carried out at several points in time to compare the results of the individual waves (focus on transformation processes).

Representativeness of the data

Representativeness means that a sample (as good as possible) should be a complete but smaller reflection of the basic population. Therefore, the sample should also include all relevant characteristics of the whole population and accurately reflect it.

Confounding variable (e. g. role models)

A confounding factor occurs when a third variable exists that is related to both, the independent and the dependent variable. This often leads to a biased estimate of a direct relationship.

The scientific experiment

Only experimental research designs allow conclusions to be drawn about cause-effect relationships (causality). Researchers vary a situation so that the participants experience it either in one way or another. Participants are randomly assigned to different experimental conditions (= randomisation). Despite (= trotz) the variations of the different experimental conditions, the procedure must be the same across all participants (must be kept constant).

Randomisation

Random assignment to conditions is a procedure that ensures that all participants have the same chance to be assigned to one of the different conditions of an experiment. This procedure guarantees that individual differences, for example the personality or socio-economic background of a participant, are evenly distributed amongst the experimental conditions. This way, differences in the means of the dependent variable can be attributed to the different experimental conditions. One of the biggest mistakes is a general differentiation between the sexes.

A/B testing (aka split testing)

- a simple controlled experiment (e. g. common user experience (UX))
- comparison of two versions of a website or app to determine which one performs better (one version will be randomly displayed to users, so one will get version A and the other version B)
- statistical analyses are used to determine the performance of versions A and B in terms of various indicators such as the conversion rate (ratio of visitors to transactions)

Between-subjects vs within-subjects designs

- Between-subjects design:
Every participant is assigned to only one factor level of the independent variable.
- Within-subjects design:
The same participant consecutively takes part in more than one experimental condition.

Common types of research publications

- Original research article:
Article that is reporting original research about new data or theories that have not been previously published. That might be the results of new experiments, or newly derived models or simulations.
 - Review/Survey article:
Article that summarizes the current state of research on a topic. A review article surveys and summarizes previously published studies, rather than reporting new facts.
 - Meta-analysis article:
Statistical technique for combining findings from multiple independent studies on the same topic to allow for a quantitative summary of overall effects and overall trends.
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Human vision

The retina of the eye is a complex layer of many neurons and photoreceptive cells. Four neuron layers within the retina perform initial image processing on the stimulations resulting from individual photoreceptors (cones for colour perception and rods for intensity perception).

The eyes are only one part of visual processing. The brain plays an important role in processing visual input from the eyes and many brain areas as well. The brain is important for interpreting the visual information and putting this information into a meaningful context.

The reason why human vision is special is, because we can recognise familiar faces despite showing different emotional expressions and seeing them in different angles.

- Human colour vision:
Individual rods and cones by themselves do not provide information due to limitations of the optic nerve. They only respond to fixed wavelength ranges (one cell does not provide colour information). Only through combinations of photoreceptor stimuli intensity and colour descriptions can be obtained.
- Selective attention:
We are very good at focusing our attention on a particular object in our environment and disregard unimportant details.
- Human visual perception:
Holistic face processing explains that faces are not treated as a combination of different parts, but faces are processed as wholes.
- Colour constancy:
The blue/black dress problem can be explained with a phenomenon called colour constancy which is the way that our brains interpret colours. What you see in the picture depends on your individual perception and where you see it (e.g. shadows are interpreted differently by our visual system, when the shadows are removed, then the colour is perceived differently).
- Why do we see optical illusions?
The brain has developed a rulebook of how objects should look from past experiences. When interpreting new information, our brain relies on previous knowledge. The brain takes a shortcut to focus on important aspects. Optical illusions fool our brains by taking advantage of these shortcuts.

Problems with computational visual perception

Generative Adversarial Networks (GANs) revealed a new challenge for image classification: Adversarial Images. Adversarial images are images whose class category looks obvious to a human, but causes massive failures in a deep network. With only a minor distortion (= Verzerrung), a deep network's classification of the image goes from a panda to a gibbon.

A biological system saves a lot of computation through selective attention and an opportunistic sampling of visual patterns. Instead of a serial image-processing pipeline, most biological vision systems involve a tight feedback loop in which orienting and tuning of the visual sensor plays an essential role.

Hearing and speech

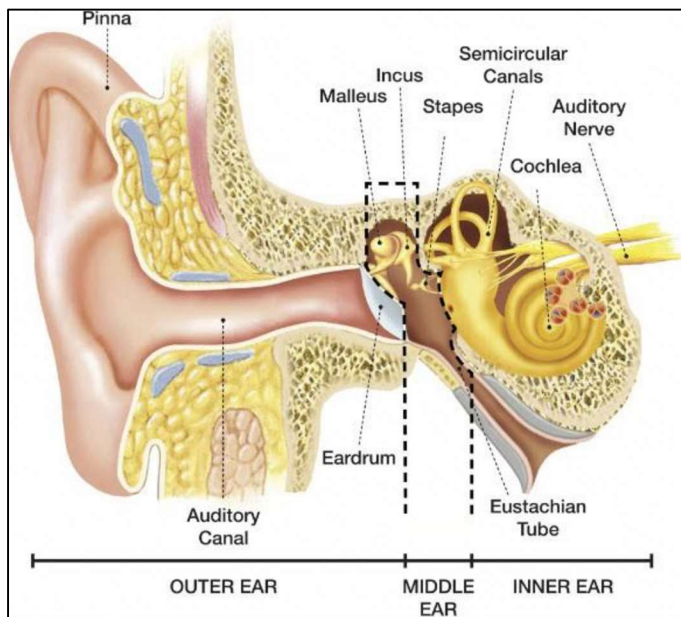
The human brain remains the most admirable speech processing machine. It works much better than computer-based speech processing and will probably continue to do so for a long time to come because the exact processes of speech processing in the brain are still largely unknown.

Human audition (= Gehör)

Sound waves are caused by the changes in pressure caused by vibrating air molecules and have 2 features, amplitude and frequency. Amplitude is the size of the wave and is related to the number of air molecules that are vibrating in a soundwave. Frequency is the rate at which sound waves vibrate. Humans cannot hear all frequencies out there.

In addition, there are two pathways in which auditory input is divided into. Those are the auditory ventral stream for identification of sound and the auditory dorsal stream for determination (= bestimmen) of how we react to sound.

- Anatomy of the outer ear:
The pinna collects and amplifies (= verstärkt) soundwaves. After the soundwaves have passed the auditory canal, they hit the eardrum.
- Anatomy of the middle ear:
The eardrum vibration sets up a vibration in the ossicles, which consists of 3 bones, malleus, incus and stapes.
- Anatomy of the inner ear:
In the inner ear, soundwaves get transduced (= übertragen) into action potentials and the auditory nerve carries signals to the brain.
- Anatomy of the cochlea:
The cochlea contains fluid within the fluid is the basilar membrane which has hair cells attached to it. Above the basilar membrane is the tectorial membrane.



Anatomy of the ear: Summary

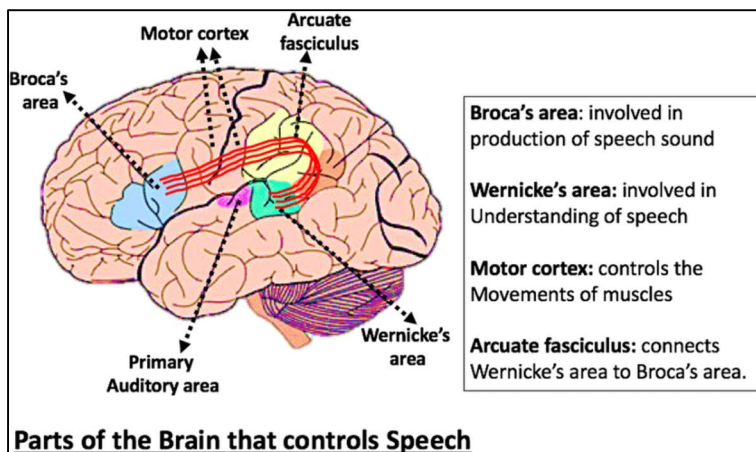
- pinna: collects and amplifies the sound waves
- auditory canal: the soundwaves from the pinna are deflected into the auditory canal
- eardrum: vibrates and sends the vibration onto the ossicles
- ossicles: little bones (hammer, anvil & stirrup) that amplify soundwaves
- cochlea: contains lymphatic fluid, within this fluid is the basilar membrane which has hair cells attached to it

Human auditory processing

- ventral stream: for speech comprehension (= Verständnis)
- dorsal stream: for sensorimotor (= Sensorik) integration (mostly dominant)

Human speech processing and understanding

Most sounds are a complex mixture of frequencies and we have a more complex neural circuitry specifically for processing speech and music.



Talking to machines as a daily routine

United Nations report that more people will speak to a voice assistance machine than to their partners in the next five years. And most of the consumers prefer a female gendered voice assistant.

„Female“ over-representation in voice assistants

All „big four“ assistants (Siri from Apple, Google Now from Google, Cortana from Microsoft and Echo from Amazon) have female voices.

Artificial voice preferences

Most companies that produce automated voices hold auditions for voice actors and collect recordings of them speaking. Then they invite focus groups to rate the voices on how well they convey certain attributes such as warmth, friendliness, competence, etc.

Some studies suggest that female synthetic voices are preferred as they are perceived as warmer compared to male voices, some studies revealed the opposite, and others come to conclusion that male voices are perceived as more intelligent.

Q – The first genderless voice

It is important to discourage gender-based insults and abusive language by genderless voices. This first gender-neutral artificial voice was created to reduce gender bias in AI assistants, and it is between 145-175 Hz (gender-neutral range according to research).

What makes human hearing special?

○ Cocktail Party Effect:

The brain is able to focus the auditory attention on a particular stimulus. Simultaneously, it is filtering out other auditory stimulus. An example for this is hearing your own name in a noisy room.

Historically, systems struggle to filter out background noise to recognise the "main speakers" voice. However, Google Research tackled this problem and trained "multi-stream convolutional neural network". When trained the system is now capable of focusing on a single voice and filtering out everything else. This means that a trained system might be even better at filtering out noises compared to a human.

What makes human communication special?

○ Understanding prosody (= Rhythmus), intonation (= Tonfall) and emotions:

Expression of emotions through prosodic modulation of the voice, in combination with other communication channels, is crucial for affective and attentional regulation in social interactions in adults and infants. Human speakers can put words and their meaning into context and can assess gestures and facial expressions simultaneously with spoken words.

Problems related to AI voice assistants

In comparison to humans, AI assistants have difficulties understanding prosody, intentions and humour from spoken language. This can cause misinterpretations of spoken language.

You could do all the machine learning in the world on the spoken word, but sarcasm is often in tone and not in word, or facial expressions. In addition, sarcasm has a lot of nonverbal cues (= Hinweise).

Human touch

- Human somatosensory system:
This system can feel hot, cold, pain, pressure, itch, vibrations, etc.
- Somatosensation can be split up into 3 sensory systems:
Those 3 systems are hapsis or touch, nociception for temperature and pain, as well as proprioception for body awareness.
- Somatosensory perceptors:
The two perceptors are skin for hapsis and nociception as well as muscles for proprioception.

Skin layers

- First layer:
Is the Epidermis, this is the layers we can see, it is made of dead skin cells and serves as a protector for the underlying layers.
- Second layer:
Is named Dermis, it contains hair follicles, blood vessels, nerve endings and touch receptors.
- Last layer:
Is the Hypodermis. It is composed of fat and connective tissue (= Gewebe). The layer of fat helps to regulate the body's temperature and the connective tissue keeps the skin attached to muscles and tendons (= Sehnen) underneath.

Types of neurons

- Proprioceptive and haptic neurons:
Carry information about proprioception (movement & location) and hapsis (touch & pressure) and is fast.
- Nociceptive neurons:
Carry pain and temperature information and is slow.

Somatosensory Pathways

There are 2 groups of neurons that travel together through the brain, the first group is for pressure and fine touch and the second group is for pain and temperature.

- Dorsal column nuclei:
Relay fine touch and pressure information. This information travels up through dorsal neurons on the right (ipsilateral) side of the spinal cord.
- Ventral spinothalamic tract:
Receives input from pain & temperature neurons. Information travels up the left side of the spinal cord.
- Medial lemniscus:
Is the pathway in the brainstem that carries information to the thalamus as well as the place where the dorsal pathway meets the ventral pathway.

Phantom limb sensations and rubber hand illusion

If a limb (= Gliedmaß) is lost through an accident or amputation. The patient feels the presence of their lost limb. This happens as neurons in the brain are sending signals (inputs) to the sensory receptors of the amputated limb. This leads the patient to feel that the limb is still there, even though it is not.

Mirror therapy is one way to help with phantom limb pain. Similar to the mirror therapy for phantom limb pain, visual feedback can also be given via virtual reality to treat phantom limb pain (receiving visual feedback through games).

A similar illusion that works for many people is the rubber hand illusion. This illusion is important for prosthetics (= Prothesen).

Cyborgs: Cybernetic Organism

There are 3 criteria for being a person with organic and biomechatronic body parts:

- communication between body and brain must be intact
 - technical aspect has to be melted with the body so it becomes a body part
 - the additional body part has to improve the human's senses/capabilities
-

Social touch in interpersonal communication

Social touching is an important and frequent part of our interactions with others. It can be calming, create bonds, express concern and solidarity, reduce anxiety or provide reassurance (= Beruhigung).

When we touch, how often we touch and how we respond to touch is influenced by a wide variety of factors (e.g., personality, context, relation to the other person, gender, culture, families, social groups, etc.).

A touch can be associated with the touch of the mother in earlier days, that made us feel secure, and can even reduce racial prejudices.

Negative effects of social touch

Sometimes, social touch can also result in negative effects. It can produce anxiety, provoke irritation or anger. Our perception of the social situation and the meaning of the touch has an impact on how welcome or intrusive (= aufdringlich) the touch is. Receiving a social touch in a competitive environment, may be seen as a sign of dominance.

Touched by robots:

Various findings suggest that a touch by a robot can enhance (= verbessern) positive feelings towards the robot, reduce stress and increased compliance (= Akzeptanz). But others suggest that an empathic touch by a robot is not necessarily perceived positively. In addition, people responded significantly more favourably to the instrumental touch (a touch to serve) than to the affective touch (a touch to provide comfort).

Self-determination theory

Is a meta-theory on motivation that, so on the reasons what makes you act. This theory differentiates between autonomous motivation and controlled motivation:

- Autonomous motivation (intrinsic):
Doing something because you feel a full sense of willingness, choice, interest, or enjoyment.
- Controlled motivation (extrinsic):
Doing something because you are demanded to do it, to get some reward or to avoid punishment.

The SDT includes the following 6 mini-theories:

- basic psychological needs theory (BPNT)
- cognitive evaluation theory (CET)
- causality orientations theory (COT)
- organismic integration theory (OIT)
- goal contents theory (GCT)
- relationships motivation theory (RMT)

Basic psychological needs theory (BPNT)

The motivation to do a task depends on the satisfaction (via the task) of the three psychological needs: Autonomy, Competence and Relatedness. Frustration of the needs leads to demotivation. Only satisfaction of all three needs leads to autonomous motivation.

- AUTONOMY as basic psychological need:
Autonomy refers to the experience of volition and willingness. When satisfied, one experiences a sense of integrity as one's actions, decisions, thoughts, and feelings are volitional (= willentlich) and authentic. When frustrated, one experiences a sense of pressure and often conflict, such as feeling pushed in an unwanted direction.
 - COMPETENCE as basic psychological need:
Competence concerns the experience of effectiveness and mastery, to feel confident in relation to whatever you doing. It becomes satisfied as one effectively engages in activities and experiences opportunities for using and extending skills and expertise. When frustrated, one experiences a sense of ineffectiveness or helplessness.
 - RELATEDNESS as basic psychological need:
Relatedness denotes the experience of warmth, bonding, and care; to be cared for by others, to care for others, to feel like you belong to a group that is important to you. It is satisfied by connecting to and feeling significant to others. Relatedness frustration comes with a sense of social alienation and loneliness.
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Dystopian headlines about AI

Headlines often consists of concerns about superintelligence, the fear of unemployment, putting the power of AI into the wrong people's hands, and general concern and caution when it comes to new technology or machine intelligence.

Constant fear-mongering (= Panikmache) can make it harder for the human brain to correctly distinguish (= unterscheiden) between a legitimate and a false threat (worrying about doomsday).

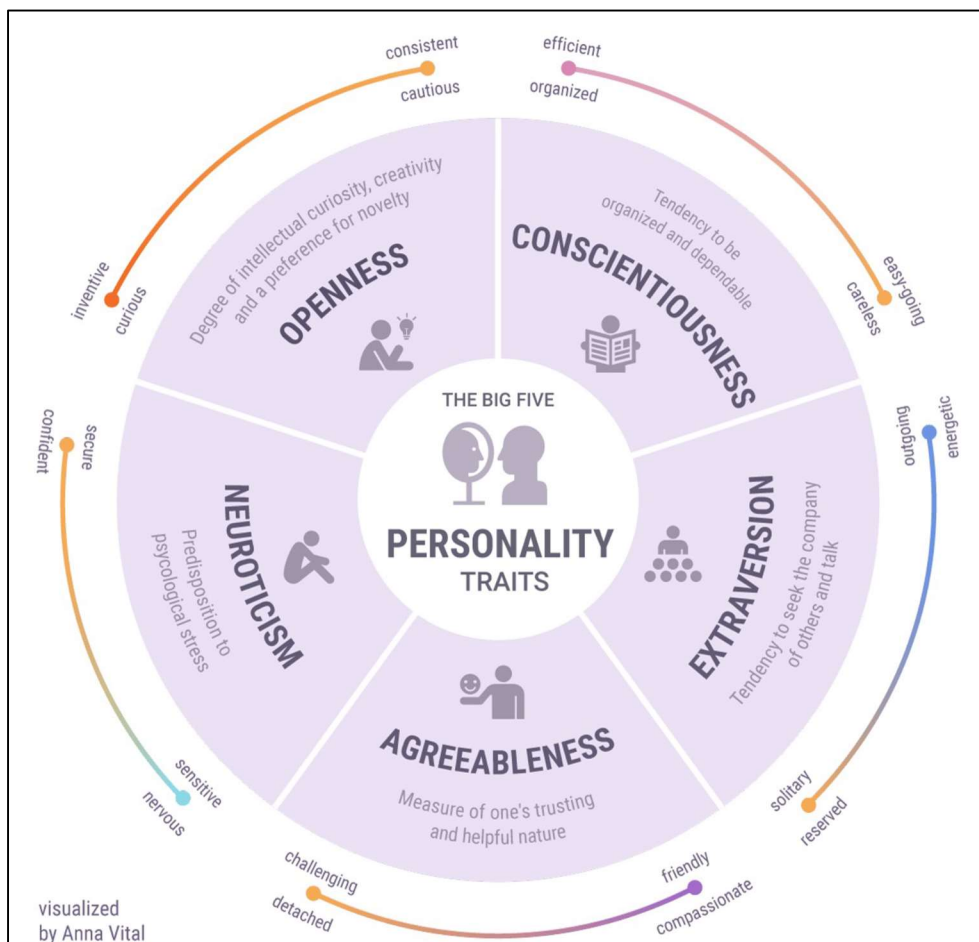
Computer anxiety

Computer anxiety is defined as the tendency of individuals to be uneasy (= unruhig), apprehensive (= besorgt), or fearful about current or future use of computers. There are computer anxiety in relation to gender, age, other anxieties, education level and educational background.

In the most cases it is about self-efficacy (= Eigenständigkeit). The definition of self-efficacy is a person's belief in his/her ability to succeed in a specific situation or accomplish a task. One's sense of self efficacy can play a major role in how one approaches goals, tasks, and challenges.

In general different factors may predict fear of AI. It is to be assumed that fear of technology results from the interaction of various influencing factors, which can relate to the user, the technology itself, to cultural or contextual factors.

The role of personality traits: Big five model



How people's personalities differ can be related to how anxious they are about new technology. Personality traits that may play a role include:

- Openness:
A person's willingness to try new things (negatively correlated with tech anxiety).
- Neuroticism:
A person's (low) emotional stability and pessimism (positively correlated with tech anxiety).

Relation between intention and actual usage: The theory of planned behavior

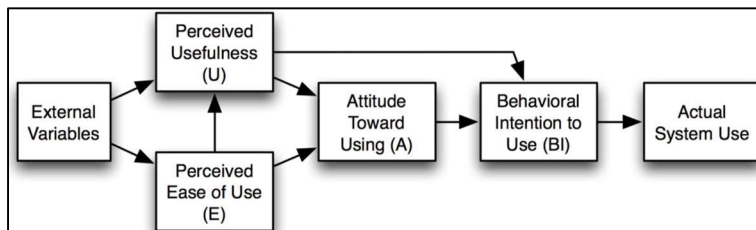
The theory states that an ...

- individual's attitude (e.g. "I think autonomous vehicles are cool"),
- perceived subjective norms (e.g. "My peers find autonomous vehicles awesome"), and
- perceived behavioral control (e.g. "I could handle the loss of control")

... together shape behavioral intentions and actual behaviors.

The technology acceptance model (TAM)

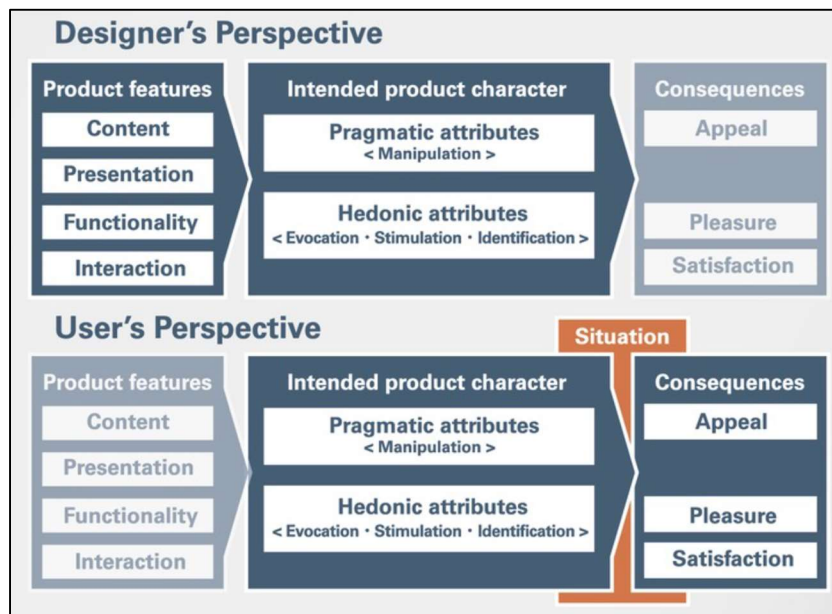
TAM focuses on the prediction of technology adoption by new users and posits that the individual adoption and use of tech is determined by perceived usefulness and perceived ease of use.



Determinants of perceived usefulness in TAM 3

- Subject Norm:
The degree to which an individual perceives that most people who are important to him/her think he/she should or should not use the system.
- Perceived Ease of Use:
The degree to which a person believes that using an IT will be free of effort.
- System Image:
The degree to which an individual perceives that use of an innovation will enhance his or her status in his or her social system.
- Job Relevance:
The degree to which an individual believes that the target system is applicable to his or her job.
- Output Quality:
The degree to which an individual believes that the system performs his or her job tasks well.
- Result Demonstrability:
The degree to which an individual believes that the results of using a system are tangible, observable, and communicable.

The model of user experience (UX)



Conclusion

For the acceptance of AI applications in society, utilitarian product attributes such as usefulness, ease of use and expected output quality are important. But basic psychological needs such as autonomy, competence and relatedness or hedonic needs such as stimulation and enjoyment should not be ignored.

Anthropomorphism

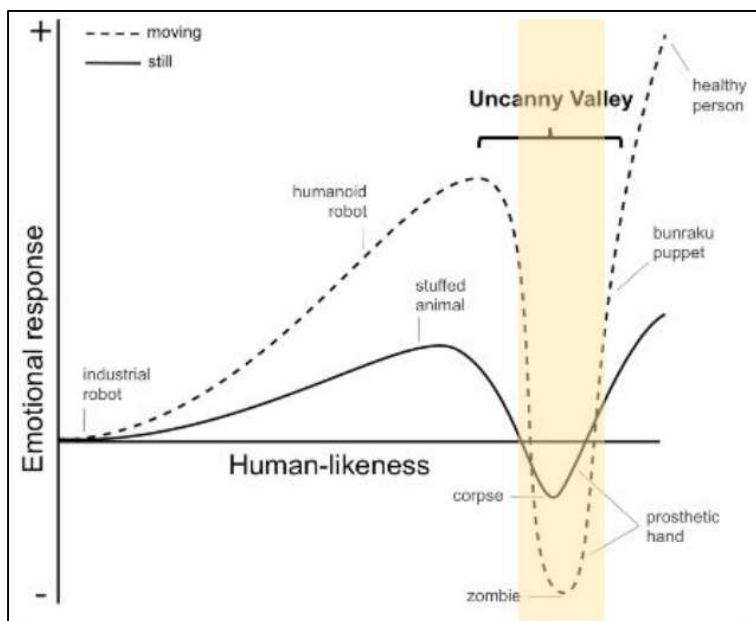
Anthropomorphism is the attribution of human traits, emotions, or intentions to non-human entities. There is a universal tendency among mankind to conceive all beings like themselves.

The Uncanny (= unheimlich) by Freud

Freud emphasized that what is uncanny is something that seems to be “un-homely” (unheimlich) and unfamiliar, but at the same time “homely” and familiar (“the unfamiliar in the familiar”). Freud’s view, the uncanny might be anything we experience in adulthood that reminds us of early psychological stages. As one of his examples, Freud names the concept of the uncanny doppelgänger (double).

The Uncanny Valley by Mori

In 1970, Mori published “Bukimi No Tani” (The Uncanny Valley). The Uncanny Valley hypothesis predicts that as an artificial figure looks more humanlike it is also perceived as more agreeable, but only until it comes to look so humanlike that we start to find its nonhuman imperfections unsettling (= seine nicht-menschlichen Unvollkommenheiten als beunruhigend empfinden). At this point, acceptance declines dramatically: the figure is regarded as uncanny / eerie / creepy. His observations led Mori to the belief that robot builders should not attempt to make their creations overly (= übermäßig) lifelike and realistic in appearance.



Proposed mechanisms behind uncanny effects

The reasons why highly but not perfectly humanlike artificial figures are creepy are; evolutionary (e.g. mate selection), categorical uncertainty (e.g. human, machine or hybrid) and wrong expectations and predictions concerning the behaviour of the robots (e.g. not human-like as expected).

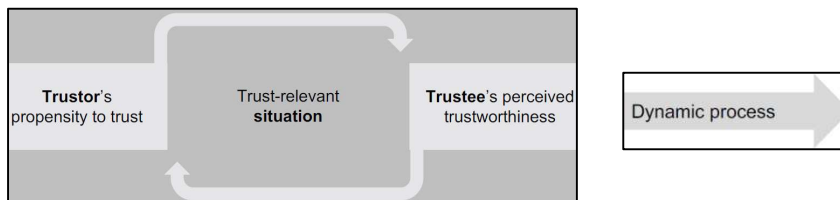
Human-Robot Trust

Trust as a basic requirement for shared spaces and close physical collaboration with robots as well as for the use of AI and algorithmic decision support systems.

Definition of Trust

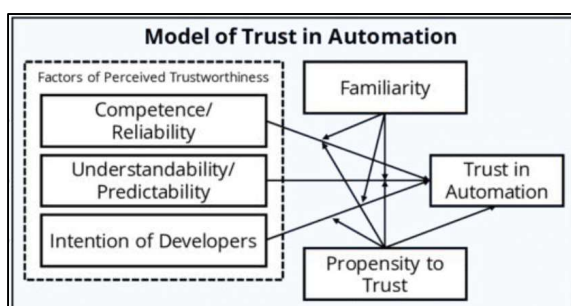
Trust is defined as “reliance on or confidence in the dependability of someone or something. In interpersonal relationships, trust refers to the confidence that a person or group of people has in the reliability of another person or group; specifically, it is the degree to which each party feels that they can depend on the other party to do what they say they will do. The key factor is not the intrinsic honesty of the other people but their predictability. Trust is considered by most psychologists to be a primary component in mature relationships with others.”

Interplay between Trustor, Trustee & Situation

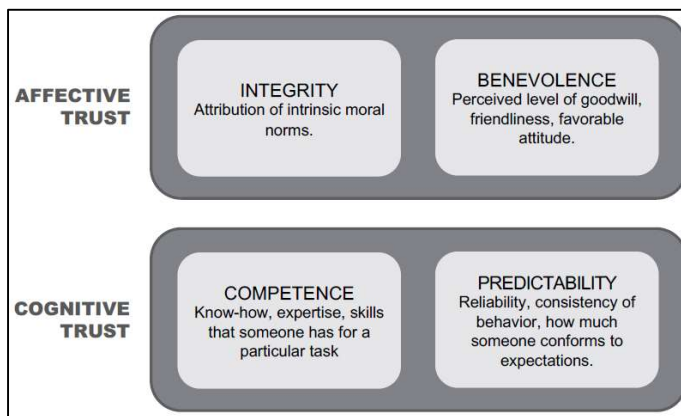


- Trustor's Propensity (= Neigung) to Trust:
The impact of trust propensity is most salient early in interpersonal interactions, when other information may not yet be available. Once other information becomes more salient, such as the trustee's previous behaviors, propensity to trust will have a weaker influence on the extent to which the trustor will make him/herself vulnerable to the trustee.
- Trustee's Perceived Trustworthiness:
Perceptions of trustworthiness are a function of the interaction of trustor and trustee as the trustor is processing information about the trustee. As interactions mature, a trustor will increasingly depend on the behavior of the trustee rather than personal factors, such as propensity to trust, when making trust evaluations.
- Trust-Relevant Situation:
Trust-relevant (or trust diagnostic) situations involve vulnerability (= Verwundbarkeit) and require some stake (i.e., risk), be it social, financial, personal, or organizational in nature. Without being vulnerable or at risk, there is no need for trust.

Theoretical Model on Trust in Automation



Theoretical Foundations: Dimensions of Trust



- **Affective/Social Trust:**
Affective trust is the confidence one places in a partner on the basis of feelings generated by the level of benevolence/care and integrity the trustee demonstrates. Affective trust is decidedly more confined to personal experiences with the focal partner than cognitive trust. As emotional connections deepen, trust in a partner may venture beyond that which is justified by available knowledge. This emotion-drive element of trust makes the relationship less transparent to objective risk assessments.
- **Cognitive Trust:**
Cognitive trust describes the willingness to rely on a partner's ability/competence and predictability/reliableness. Cognitive trust is knowledge-driven, the need to trust presumes a state of incomplete knowledge. A state of complete certainty regarding a partner's future actions implies that risk is eliminated and trust is redundant.

System Interpretability as Requirement for the Right Calibration of Trust

As autonomous systems become more complex and, the ability of human operators to understand the system decrease. Previous studies have shown that trust is positively correlated to how much users understand the functions of the system. Trust has been identified as a key factor that influences whether people rely on decisions made by autonomous systems, especially in situations of uncertainty. Facilitating appropriate levels of trust in automation is central to maximizing performance and safety of human-automation teams.

Goals of Research on Interpretability and Trust: Avoidance of Over-Trust and Under-Trust

- **Unwarranted Over-Trust:**
May lead a human operator to overestimate a system's ability's and to rely too much on them,
- **Unwarranted Under-Trust:**
May lead to reluctance or even to fear to use the system.

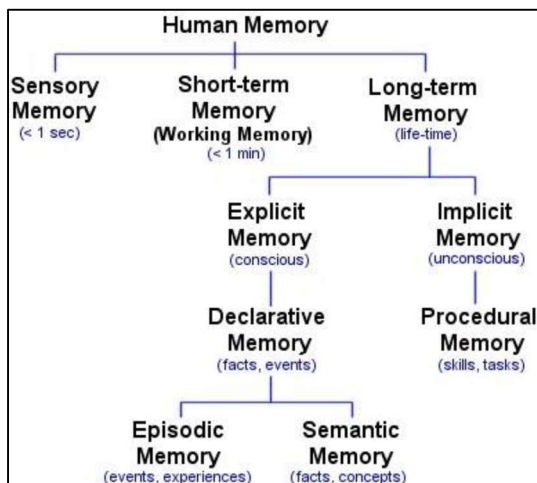
Operators/users must be given the chance to understand how a system works and thus establish an appropriate level of trust.

Memory Processes

- encoding: acquiring the memory and consolidating it
- storage: how a memory is maintained
- retrieval: how a memory is retrieved (= Abrufen)

Human Memory

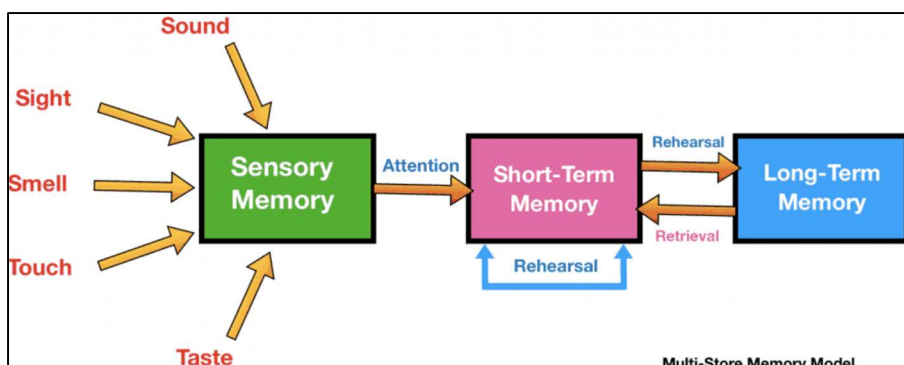
Besides Long-Term Memory (LTM) and Short-Term Memory (STM) human memory can be divided in many different sub-types of memory:



Brain Areas involved in Memory

Brain Area	Associated Memory Function
Frontal lobes:	Short-term memory tasks
Prefrontal cortex, parts of temporal lobes:	Efficient encoding of words, pictures
Hippocampus:	Formation of long-term declarative memories; may “bind together” diverse elements of a memory so it can be retrieved later as a entity
Cerebellum:	Formation and retention of simple classically conditioned responses
Cerebral cortex:	Storage of long-term memories, possibly in areas involved in the original perception of the information

Atkinson-Shiffrin Multi-Store Memory Model (1968)



Artificial Memory

- computing: input and output, information processing & memory
- match: complex movement, vision, language & structured problem solving
- brain: creativity, emotion and empathy, planning & consciousness (= Bewusstsein)

With regard to number capacity, of course an AI system is able to store much more information. But humans are able to combine the different memory types, learn from their past experiences and draw conclusions for upcoming experiences.

The “Sleeper Effect”

The so-called sleeper effect means that after a certain period recipients might only remember the content of a message but forget the source and how trustworthy the source was. This means: Even if recipients identify fake news as such and therefore initially classify them as not trustworthy, it can happen that after a few weeks they only remember the content of the report but forget that they originally thought it was untrue. They can no longer attribute the information to any source.

Positive Memory Bias

People tend to remember more pleasant than unpleasant information, a result of powerful mechanisms that maintain our well-being (e.g. remember events as more favourably than they actually were and view their past through rosy glasses).

Loftus and Palmer Experiment (1974)

Elizabeth Loftus (Psychologist) believes that a memory of an event that was witnessed (=is highly flexible, more so if this was an emotional event (negative experience)). Their study showed that language was able to alter participants memory of a car accident. Depending on how the questions were frased, participants would remember the car’s speed differently.

- The question was:
“About how fast were the cars going when they (smashed / collided / bumped / hit / contacted) each other?”

The results revealed that the estimated speed varied depending on the verbs used. The verb affected the participant’s memory of the event.

Eyewitness Testimony (= Aussagen von Augenzeugen)

Since a crime is an unplanned event and often emotional and stressful event, eye witnesses are not prepared and unable to focus on smaller details. Race bias, previous experiences, knowledge, clothing etc. can all play a role in making the wrong decision as an eye witness.

Forgetting...?

Memory loss can be caused by; Dementia (e.g. Alzheimer's disease), amnesia, head injury, stroke, stress, drug use, nutritional deficiency and sleep deprivation.

Priming-Effect: Implicit Memory vs. Explicit Memory

Priming means "to prepare", so a priming effect happens in a demonstration where a preceding stimulus influences the processing or perception of a subsequent stimulus.

Priming is linked to implicit memory as it is related to the retention of information without conscious recollection of past experiences. Exposure to a stimulus (e.g. white colour) can influence the response to a later stimulus (e.g. cow drinks milk, because milk is white).

- explicit memory: recall or recognition & conscious recollection of a past event/experience

Learning & Conditioning

There are two types of conditioning:

- Classical Conditioning (Ivan Pavlov):
Focuses on an association between two stimuli, a pre-existing stimulus (unconditioned) and a neutral one (associate an involuntary response and a stimulus).
 - Operant Conditioning (B.F. Skinner):
Focuses on punishment or reinforcement learning (associate a voluntary behaviour and a consequence).
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Consciousness Definition by Merriam- Webster

- the quality or state of being aware especially of something within oneself
- the state or fact of being conscious of an external object, state, or fact
- concern for some social or political cause (awareness)
- the state of being characterized by sensation, emotion, volition, and thought (mind)
- the totality of conscious states of an individual

The Mind-Body Problem

The "mind-body problem" tackles the question of the relationship between the mind and the body. You differentiate between the mental realm (thoughts, beliefs, pain, sensations and emotions) and the physical realm (matter, atoms, neurons).

Different Standpoints on Consciousness

- (Cartesian) Dualism:
 - Dualism is the view that the mind and body both exist as separate entities.
 - Humans can be characterized by having both a mind (nonphysical) and body (physical).
 - The mind controls the body and the body can influence the mind (= the mind and body enter in a two-way causal interaction).
 - The mind and body are able to exist independently (e.g. after death)
- Monism/ Materialism:
 - Nothing exists apart from the material world (brain = physical matter)
 - Consciousness is simply a function of the brain
 - Humans are complicated physiological organisms not more (= there are no souls or immaterial minds)

What is it like to be a Bat? (Thomas Nagel, 1974)

“But even if we can imagine all of these things, it only tells us what it is like for me to be a bat, or for me to behave as a bat behaves. It does not tell us what it is like for a bat to be a bat.”

Phenomenal Consciousness (Ned Block, 1995)

Phenomenal Consciousness is about Experience and Awareness (feeling pain, perspective of view). What it is like to feel something is dependent on the subjective properties of an experience. Not paying attention to an auditory stimulus could be a case for Phenomenal-Consciousness without Access-Consciousness.

Access Consciousness (Ned Block, 1995)

Access Consciousness is a more cognitive standpoint/ non-phenomenal category of consciousness. A state is Access-conscious if it is able to be used for the rational control of thought and action. Blindsight might be a case for Access-Consciousness without Phenomenal-Consciousness.

Mary's Room and its Message for AI Developments

Just because an AI would be able to mimic every single function of the human brain, this does NOT mean that the AI has a conscious brain.

The Hard and Easy Problems of Consciousness

- The Easy Problem of Consciousness:
 - Being asleep vs. Being awake
 - Ability to discriminate, categorize and react to environmental stimuli
 - Integration of information by a cognitive system
 - The ability of a system to access its own internal states
 - The focus of attention
 - The deliberate control of behaviour
 - The Hard Problem of Consciousness:
 - How do physical processes in the brain give rise to the subjective experiences of the mind?
 - Subjective point of view vs. Physical mechanism
 - When we see, we see colour, shapes, and we experience these from our point of view and this is the conscious experience of seeing „inner movie of the mind“.
 - “There is something it is like to be a conscious organism.” (Nagel, 1974)
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